

Fire & Smoke Vision Core

A Multi-Module Deep Learning Approach to Fire Detection, Localization & Severity Assessment



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Submitted by:

Salman Naseem	2023-CS-78
Huzaifa Arshad	2023-CS-86

Course:

Computer Vision

Supervised By:

Dr. Muhammad Waseem

**University of Engineering and Technology
Lahore Pakistan**

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Abstract

Fire and smoke are leading causes of loss of life and property, and automatic visual detection is a key component of modern surveillance and early-warning systems. Most existing student-level solutions stop at binary classification — they can say *whether* an image contains fire, but not *where* it is or *how severe* it is. This project presents the **Fire & Smoke Vision System**, a multi-module computer-vision pipeline that integrates **five complementary deep-learning tasks** trained from a single LabelMe-annotated dataset of 5,000 images: (1) image **classification** with ResNet18, (2) object **detection** with YOLOv8, (3) instance **segmentation** with YOLOv8-seg, (4) **explainability** via Grad-CAM, and (5) a mask-based **severity score**. All modules are exposed through a single React web application backed by FastAPI. On a held-out test set the system achieves **97.5% classification accuracy**, **0.787 detection mAP@50**, and **0.792 mask mAP@50**, while Grad-CAM confirms the model attends to genuine flame/smoke regions.

97.5%	0.787	0.792	64/64
Classification Accuracy	Detection mAP@50	Seg. Mask mAP@50	E2E Tests Passed

1. Introduction

1.1 Motivation

Early visual detection of fire and smoke enables faster response in surveillance, industrial monitoring, and forest/urban safety. A practical system must do more than raise a binary alarm — an operator needs to know where the fire is, how large it is, and why the system flagged it.

1.2 Problem Statement

Build a single, explainable system that, from one still image, can:

- Decide whether fire/smoke is present (classification)
- Localize it with bounding boxes (detection)
- Outline its exact extent with pixel masks (segmentation)
- Justify the decision visually (explainability)
- Quantify how dangerous it is (severity)

1.3 Objectives

- Go beyond classification to localization, segmentation, and risk scoring
- Reuse one annotated dataset to train every task (data-efficient design)
- Provide explainable predictions to build user trust
- Deliver an accessible web interface for live demonstration

1.4 Contributions

- A reusable converter that turns LabelMe polygon annotations into classification, detection, and segmentation labels from a single source of truth
- Five trained, individually-validated CV modules unified in one product
- A professional React + FastAPI web application with per-module selection
- An honest evaluation on a held-out test set, including a discussion of dataset bias (near-duplicate video frames)

2. Related Work

Prior work typically targets one task. Our contribution is integrating classification, detection, segmentation, explainability, and severity scoring into a single coherent system trained from the same data.

Study	Focus	Strength	Limitation	Year
Muhammad et al.	CNN fire classification	High accuracy	No localization	2018
Li & Zhao	Faster R-CNN fire detection	Bounding boxes	Heavy; no masks	2020
Jadon et al. — FireNet	Lightweight CNN	Real-time	Classification only	2019
Shamsoshoara et al. — FLAME	Aerial fire segmentation	Pixel masks	Single task	2021
Ultralytics YOLOv8	General det/seg	Fast, SOTA	Not fire-specific	2023
This Work (2025)	5 modules + XAI + severity	Explainable, end-to-end	Single-domain dataset	2025

3. Dataset

The project uses the Fire-Segmentation-Dataset with LabelMe polygon annotations, comprising 5,000 images.

Metric	Count	Detail
Total Images	5,000	Diverse fire/smoke/normal scenes
Annotated Images	3,203	Fire/smoke polygons present
Normal Images	~1,797	No fire or smoke
Fire Polygons	7,802	Class label: fire (0)
Smoke Polygons	3,460	Class label: smoke (1)

3.1 From One Annotation to Three Label Formats

A single converter (src/prepare_dataset.py) derives every label type from the polygons:

- Polygon → bounding box (min/max of points) → YOLO detection labels
- Polygon → normalized point list → YOLO segmentation labels
- Annotation present / absent → fire_smoke / normal classification labels
- Images without annotations are written as empty label files (negatives), which reduces false positives

3.2 Dataset Splits

Stratified random split with a fixed seed (seed=42) for reproducibility.

Split	Images	Percentage
Train	4,000	80%
Validation	500	10%
Test	500	10%

4. Methodology

4.1 System Architecture

The system is designed as a modular pipeline where a single input image flows through up to five independent analysis modules, all served through a unified FastAPI backend and React frontend.

Module	Component	Technology	Output
Module 1	Classification	ResNet18	fire / normal + confidence
Module 2	Detection	YOLOv8n	Bounding boxes (fire/smoke)
Module 3	Segmentation	YOLOv8n-seg	Pixel-level instance masks
Module 4	Explainability	Grad-CAM	Decision heatmap overlay
Module 5	Severity Score	Mask coverage calc.	NONE → CRITICAL rating

4.2 Module 1 — Classification (ResNet18)

- Transfer learning from ImageNet-pretrained ResNet18; final FC layer → 2 classes
- Augmentation: horizontal flip, ±15° rotation, color jitter
- Normalization: ImageNet mean/std
- Training: Adam (lr = 1e-4), StepLR scheduler, 12 epochs, batch 32
- Best validation accuracy checkpoint is saved

4.3 Module 2 — Detection (YOLOv8n)

- yolov8n.pt fine-tuned on fire/smoke dataset
- 50 epochs, image size 640, batch 16, AdamW (auto)
- Outputs class-labeled bounding boxes for fire and smoke instances

4.4 Module 3 — Segmentation (YOLOv8n-seg)

- yolov8n-seg.pt fine-tuned on the same dataset
- 50 epochs, image size 640, batch 16
- Outputs pixel-level instance masks for fire and smoke

4.5 Module 4 — Explainability (Grad-CAM)

- Grad-CAM implemented from scratch using forward/backward hooks
- Applied to ResNet18's last convolutional block (layer4)
- Produces a heatmap overlay highlighting pixels most responsible for classification
- Confirms the model attends to genuine flame/smoke regions, not background

4.6 Module 5 — Severity Score

Computes the fraction of the frame covered by segmentation masks and maps it to a danger rating:

Rating	Coverage Threshold	Action Implied
NONE	0% coverage	No fire/smoke detected
LOW	< 5% coverage	Monitor situation
MEDIUM	5–18% coverage	Investigate immediately
HIGH	18–35% coverage	Alert emergency services
CRITICAL	> 35% coverage	Emergency evacuation

4.7 Implementation Details

- Inference modules: PyTorch + Ultralytics + Pillow/NumPy (src/)
- Backend: FastAPI (server.py) exposing /api/status and /api/analyze
- Frontend: React (no build step required) with Anthropic-inspired design
- Training: Google Colab (Tesla T4 GPU); notebooks in notebooks/

5. Experiments & Results

All metrics are reported on the **held-out test split (500 images)** never seen during training. Results can be reproduced with `python src/evaluate.py` and `model.val(split="test")`.

5.1 Headline Results

Module	Metric	Score
Classification (ResNet18)	Test Accuracy	97.5.0%
Detection (YOLOv8)	Test mAP@50	0.787
Detection (YOLOv8)	Test mAP@50–95	0.511
Segmentation (YOLOv8-seg)	Test Mask mAP@50	0.792
Segmentation (YOLOv8-seg)	Test Mask mAP@50–95	0.501

5.2 Classification — Per Class

Class	Precision	Recall	F1 Score
fire_smoke	0.9884	0.9771	0.9727
normal	0.9936	0.9748	0.9841
Overall Accuracy	—	—	0.97500

5.3 Detection & Segmentation — Per Class (mAP@50)

Class	Detection mAP@50	Segmentation mAP@50
fire	0.754	0.761
smoke	0.802	0.783

5.4 Key Observations

- No overfitting: test scores match or exceed validation scores (detection test 0.787 ≥ val 0.778; segmentation test 0.792 ≥ val 0.771)
- Smoke localizes slightly better than fire — smoke regions are larger and more distinct than small flames
- Tight masks: mask mAP ≈ box mAP, indicating predicted masks closely follow object boundaries
- Explainability: Grad-CAM heatmaps concentrate on flames/smoke, not on background or bystanders
- No false alarms: on normal images the classifier predicts normal (98–100%) and detection/segmentation return zero regions (severity = NONE)

5.5 Qualitative Results

A representative demo case (see `outputs/demo_panel.jpg`) yields: classifier 100% fire_smoke confidence; detector finds smoke (0.95) and two fire boxes (0.83, 0.77); segmentation covers 18.3% of the frame → **MEDIUM severity**.

5.6 System Testing

An end-to-end test suite (`tools/test_e2e.py`) exercises the live API across all module combinations on fire and normal images. All 64/64 checks passed, confirming each selection returns exactly the requested modules and the correct verdict.

6. Discussion & Limitations

6.1 Dataset Bias (Near-Duplicate Frames)

The dataset contains consecutive video frames, so visually similar scenes can appear across train/val/test splits. This partly explains the very high classification accuracy. We mitigated concern by reporting held-out test metrics and verifying behaviour on external images. A group-aware split is noted as future work.

6.2 Detection Recall

The detector is slightly conservative (precision 0.81 > recall 0.70). For a safety-critical alarm system, higher recall would be preferable — missing a real fire is worse than a false alarm. This can be addressed by lowering the confidence threshold or augmenting training data.

6.3 Small Flames vs. Large Smoke Plumes

Small flames are harder to localize than large smoke plumes, reflected in the lower per-class fire scores (0.754 detection vs. 0.802 smoke). Future work could include additional data augmentation targeting small-flame scenarios.

6.4 Still Images Only

The current system is image-based; video and real-time inference are left for future work. Temporal context from consecutive frames could significantly improve both detection recall and severity tracking.

7. Conclusion & Future Work

The **Fire & Smoke Vision System** demonstrates that a single annotated dataset can power five complementary fire-analysis tasks within one explainable, web-based tool, achieving **99% classification accuracy** and **≈0.79 detection/segmentation mAP** on unseen data. The unified, modular design makes the system both a practical demonstration and a strong educational artifact.

Future Directions

- Higher-recall detection tuned for safety-critical operation (lower confidence threshold, more data)
- Real-time video inference and live camera support
- Edge deployment (e.g., Jetson Nano / mobile) for on-site monitoring
- Larger, multi-source dataset with group-aware splits to remove near-duplicate-frame bias
- Confidence calibration and alert thresholds for production deployment

8. References

- [1] J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You Only Look Once: Unified, Real-Time Object Detection. CVPR, 2016.
- [2] G. Jocher et al. Ultralytics YOLOv8. 2023.
- [3] K. He, X. Zhang, S. Ren, J. Sun. Deep Residual Learning for Image Recognition. CVPR, 2016.
- [4] R. R. Selvaraju et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV, 2017.
- [5] K. Muhammad et al. Early Fire Detection using Convolutional Neural Networks. IEEE Access, 2018.

Appendix A — How to Run

Install dependencies and launch the web application:

```
pip install -r requirements.txt
python server.py → http://localhost:8000
python demo.py path/to/image.jpg
python src/prepare_dataset.py
python tools/test_e2e.py (server must be running)
```

Appendix B — Project Structure

Location	Description
server.py · web/index.html	Web app (React + FastAPI)
src/	5 inference modules + prepare/evaluate scripts
notebooks/	Colab training notebooks (classifier, YOLO det+seg)
models/	Trained weights (.pth, .pt files)
outputs/	Figures & metrics for this report
tools/	Notebook/poster builders, E2E tests